Signatures simplified

Let $X_t = (X_t^1, \ldots, X_t^d)$ be a continuous semimartingale. Then the $(i_1, \ldots, i_n)$th component of the order-n part of the signature $\hat{\mathbb{X}}_t$ of X_t is

$$\hat{\mathbb{X}}_t^{(i_1,\ldots,i_n)} \quad = \quad \int_{u_n=0}^{t} \int_{u_{n-1}=0}^{u_n} \cdots \int_{u_1=0}^{u_2} dX_{u_1}^{i_1} \circ \cdots \circ dX_{u_n}^{i_n}.$$

Here $\int_0^t Y_s \circ dX_s := \int_0^t Y_s dX_s + \frac{1}{2}[X,Y]_t$ is the **Stratonovich** integral of Y with respect to X, and the final term is the quadratic covariation of X and Y. This ensures that $X_t Y_t = X_0 Y_0 + \int_0^t Y_s \circ dX_s + \int_0^t X_s \circ dY_s$ i.e. Stratonovich integration obeys the usual rules of calculus.

We will generally be interested in the case $X_t = (t, W_t)$, which we call *time-augmented* Brownian motion. In this case we denote $\hat{\mathbb{X}}$ by $\hat{\mathbb{W}}$, so $d[W,W]_t = dt$, $d[t,W]_t = d[W,t]_t = 0$, and $d[t,t]_t = 0$.

Expectation of the signature of (t, W_t): the Fawcett formula

Let $i_k \in \{1, 2\}$ for $k = 1, \ldots, n$, where 1 refers to the time dimension and 2 to the spatial, i.e. W, dimension. Let x denote the number of time-dimension terms in $(i_1, \ldots, i_n)$, i.e. the number of 1s. Then the $(i_1, \ldots, i_n)$th component of the level-n part of $\mathbb{E}(\hat{\mathbb{W}}_T)$ is

$$\frac{T^{\frac{1}{2}(n+x)}}{(\frac{1}{2}(n+x))!\; 2^{\frac{1}{2}(n-x)}} \tag{1}$$

if the word is a concatenation of blocks 1 and 22, and otherwise the expectation is zero. Equivalently, either $x = n$, or every occurrence of 2 belongs to a consecutive pair 22 in this block decomposition. For example, 122 and 221 are allowed, but 212 and 222 are not.

We list the non-zero components of $\mathbb{E}(\hat{\mathbb{W}}_T)$ from levels 2 to 4 here:

$$\begin{aligned} \mathbb{E}(S^2) &= \frac{T^2}{2}(11) + \frac{T}{2}(22) \\ \mathbb{E}(S^3) &= \frac{T^3}{6}(111) + \frac{T^2}{4}((122),(221)) \\ \mathbb{E}(S^4) &= \frac{T^4}{24}(1111) + \frac{T^3}{12}((1122),(1221),(2211)) + \frac{T^2}{8}(2222). \end{aligned} \tag{2}$$

The level-0 component is 1, and the level-1 component is $T(1)$. A simple Monte Carlo test of these formulae can be performed using the `iisignature` package with antithetic sampling.

Example: let $n = 3$. Then the $(1,2,2)$th component of $\hat{\mathbb{W}}$ corresponds to $x = 1$ and has a pair of 2s, so the expectation computed using Eq. (1) is $\frac{1}{4}T^2$. To check this manually from the definition of the signature, we compute

$$\begin{aligned} \int_0^T \int_0^t \int_0^s du \circ dW_s \circ dW_t &= \int_0^T \int_0^t s \circ dW_s \circ dW_t \\ &= \int_0^T (\int_0^t s\, dW_s) dW_t + \frac{1}{2}\int_0^T t\, dt. \end{aligned}$$

The Itô integral has expectation zero, so this component has expectation $\frac{1}{4}T^2$.

Demystifying the shuffle product: expressing products as linear functionals of $\hat{\mathbb{W}}_t$

Consider the simple product of the coordinate functionals $\langle e_1^*, \hat{\mathbb{W}}_t \rangle = t$ and $\langle e_2^*, \hat{\mathbb{W}}_t \rangle = W_t$. Comparing this to the off-diagonal terms of the level-2 signature, $\int_0^t W_s ds$ and $\int_0^t s \circ dW_s = tW_t - \int_0^t W_s ds$, we see that their sum is equal to tW_t, and hence

$$\langle e_1^*, \hat{\mathbb{W}}_t \rangle \langle e_2^*, \hat{\mathbb{W}}_t \rangle \quad = \quad tW_t \;=\; \langle e_1^* \sqcup\!\sqcup\, e_2^*, \hat{\mathbb{W}}_t \rangle.$$

In this case $e_1^* \sqcup\!\sqcup\, e_2^*$ has level-2 terms $(12) + (21)$ and is zero elsewhere. This is still a linear functional of $\hat{\mathbb{W}}_t$, although $\hat{\mathbb{W}}_t$ itself contains non-linear terms. In this case we have two decks of one card each, so the shuffle product consists of just the two words 12 and 21.

Now consider $\int_0^t W_u du \cdot W_t$. This is the product of the signature component corresponding to the word $(2,1)$ and the component corresponding to the word (2). We wish to rewrite it as a linear functional of level-3 signature terms. We first write

$$\int_0^t W_u du \cdot W_t \quad = \quad \int_0^t \int_0^u \circ dW_r\, du \cdot \int_0^t \circ dW_s \;=\; \int_0^t \int_0^t \int_0^u \circ dW_r \circ dW_s\, du.$$

We can break this up as

$$\int_0^t \int_0^u \int_0^u \circ dW_r \circ dW_s\, du + \int_0^t \int_u^t \int_0^u \circ dW_r \circ dW_s\, du. \tag{3}$$

The first integral in (3) can be written as

$$\int_0^t \int_0^u \int_0^s \circ dW_r \circ dW_s\, du + \int_0^t \int_0^u \int_s^u \circ dW_r \circ dW_s\, du, \tag{4}$$

and the final integral here can be rewritten as

$$\int_0^t \int_0^u \int_0^r \circ dW_s \circ dW_r\, du.$$

The second integral in (3) can be written as

$$\int_0^t \int_0^s \int_0^u \circ dW_r\, du \circ dW_s.$$

Adding the three blue terms, we obtain the shuffle product

$$(2,1) \sqcup\!\sqcup (2) = 2 \cdot (2,2,1) + (2,1,2).$$

Similarly, the functional $\int_0^t u\, dW_u \cdot W_t$ corresponds to the words $(1,2)$ and (2), so the shuffle product is

$$(1,2) \sqcup\!\sqcup (2) = 2 \cdot (1,2,2) + (2,1,2).$$

The proof of the shuffle formula is given in Lemma 22.2 of the 1994 article of Gaines, “The algebra of iterated stochastic integrals”.

Gaussian Volterra processes as a linear combination of signature elements

See, for example, Section 4.3 of [AGH24]. For a Gaussian Volterra process $Z_t = \int_0^t K(t-s) dW_s$, if the Taylor expansion of K about t is valid on the interval of integration, then

$$\begin{aligned} Z_t \quad &= \quad \langle \ell_t, \hat{\mathbb{W}}_t \rangle \;=\; \int_0^t \sum_{n=0}^{\infty} K^{(n)}(t) \frac{(-s)^n}{n!} dW_s \\ &= \quad \sum_{n=0}^{\infty} K^{(n)}(t) \int_0^t \frac{(-s)^n}{n!} dW_s. \end{aligned}$$

Thus Z_t is represented, at this fixed time t, as an infinite linear combination of signature terms of $\hat{\mathbb{W}}_t$. More specifically, the nth term in the series is a multiple of the level-$(n+1)$ signature term

$$\int_{s=0}^t \int_{u_n=0}^s \int_{u_{n-1}=0}^{u_n} \cdots \int_{u_1=0}^{u_2} du_1 \cdots du_n\, dW_s.$$

Here the functional is denoted by ℓ_t because its coefficients depend on the fixed terminal time t.

In particular, for the Riemann–Liouville process $Z_t = \int_0^t (t-s)^{H-\frac{1}{2}} dW_s$, the formal binomial expansion gives

$$Z_t \quad = \quad t^{H-\frac{1}{2}} \sum_{n=0}^{\infty} t^{-n} (\frac{1}{2} - H)^{\bar{n}} \frac{1}{n!} \int_0^t u^n dW_u.$$

Here $a^{\bar{n}} = a(a+1)\cdots(a+n-1)$, with $a^{\bar{0}} = 1$. This expansion should be interpreted with care at the endpoint $s = t$; in particular, for $H < \frac{1}{2}$ the kernel is singular at zero, so this is best regarded as a formal expansion unless a suitable convergence statement is supplied.

We can numerically check this by computing the covariance of both sides, where the covariance of the right hand side involves a doubly infinite sum. A *finite* number of linear signature elements is still a semimartingale, since individual terms of $\hat{\mathbb{W}}_t$ are semimartingales. However, for $H < \frac{1}{2}$ the infinite limit is not a semimartingale, since the Riemann–Liouville process then has infinite quadratic variation.

Application to stochastic volatility models - sampling the VIX

Following slide 27 in [Ger25], let

$$\begin{aligned} dS_t &= S_t \Sigma_t dB_t \\ \Sigma_t &= \langle \sigma, \hat{\mathbb{W}}_t \rangle, \end{aligned}$$

where $B_t = \rho W_t + \bar{\rho} W_t^{\perp}$, $\bar{\rho} = \sqrt{1-\rho^2}$, and W and $W^{\perp}$ are independent Brownian motions. Then, from the shuffle formula above, $V_t := \Sigma_t^2 = \langle \sigma \sqcup\!\sqcup \sigma, \hat{\mathbb{W}}_t \rangle$, so $\mathbb{E}(V_t) = \langle \sigma \sqcup\!\sqcup \sigma, \mathbb{E}(\hat{\mathbb{W}}_t) \rangle$, and we know $\mathbb{E}(\hat{\mathbb{W}}_t)$ from (1). Hence

$$\mathrm{VIX}_0^2 \;=\; \frac{1}{\Delta} \int_0^{\Delta} \mathbb{E}(V_u) du \;=\; \frac{1}{\Delta} \int_0^{\Delta} \langle \sigma \sqcup\!\sqcup \sigma, \mathbb{E}(\hat{\mathbb{W}}_u) \rangle du.$$

The quintic model of Abi Jaber and Li uses $n = 5$, with W replaced by an OU process Y, but only uses the trivial polynomial terms Y_t^n for odd n from $\hat{\mathbb{Y}}_t$, i.e. the last coordinate of the level-n signature for odd n.

Computing VIX_t^2 using conditional expectations of $\hat{\mathbb{W}}_t$

To compute $\mathrm{VIX}_t^2 = \frac{1}{\Delta} \mathbb{E}(\int_t^{t+\Delta} V_u du \,|\, \mathcal{F}_t^W)$ for $t > 0$, we need to compute conditional expectations of signature elements. To this end, first note that, with the convention $\hat{\mathbb{W}}_t^{\emptyset} = 1$,

$$\begin{aligned} d\hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-1},1)} &= \hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-1})} dt \\ d\hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-1},2)} &= \hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-1})} \circ dW_t \\ &= \hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-1})} dW_t \,+\, \frac{1}{2} d\langle \hat{\mathbb{W}}^{(i_1,\ldots,i_{n-1})}, W \rangle_t \\ &= \hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-1})} dW_t \,+\, \frac{1}{2} \hat{\mathbb{W}}_t^{(i_1,\ldots,i_{n-2})} dt \, \mathbf{1}_{\{i_{n-1}=2\}}, \end{aligned}$$

where the final line is for $n \geq 2$. Hence, for $u \geq t$ and $n \geq 2$,

$$\begin{aligned} \frac{\partial}{\partial u} \mathbb{E}\big(\hat{\mathbb{W}}_u^{(i_1,\ldots,i_{n-1},i_n)} \,\big|\, \mathcal{F}_t^W\big) &= \mathbb{E}\big(\hat{\mathbb{W}}_u^{(i_1,\ldots,i_{n-1})} \,\big|\, \mathcal{F}_t^W\big) \mathbf{1}_{\{i_n=1\}} \\ &\quad + \frac{1}{2} \mathbb{E}\big(\hat{\mathbb{W}}_u^{(i_1,\ldots,i_{n-2})} \,\big|\, \mathcal{F}_t^W\big) \mathbf{1}_{\{i_{n-1}=2, i_n=2\}}. \end{aligned}$$

Together with the initial condition

$$\mathbb{E}\big(\hat{\mathbb{W}}_t^I \,\big|\, \mathcal{F}_t^W\big) = \hat{\mathbb{W}}_t^I, \tag{5}$$

this gives a recursive ODE for $\mathbb{E}(\hat{\mathbb{W}}_u^I | \mathcal{F}_t^W)$, which, when $t = 0$, is consistent with the Fawcett formula above. The level-one cases are obtained separately from $d\hat{\mathbb{W}}_u^{(1)} = du$ and $d\hat{\mathbb{W}}_u^{(2)} = dW_u$.

Here is a specific example:

$$\begin{aligned} d\Big(\int_{s=0}^{t} \int_{u=0}^{s} \int_{v=0}^{u} dv \circ dW_u \circ dW_s\Big) &= \int_{u=0}^{t} \int_{v=0}^{u} dv \circ dW_u \circ dW_t \\ &= \int_{u=0}^{t} u \circ dW_u \circ dW_t \\ &= \Big(\int_{u=0}^{t} u \, dW_u\Big) dW_t \,+\, \frac{1}{2} t dt. \end{aligned}$$

Integrating and taking expectations, we obtain

$$\mathbb{E}\Big(\int_{s=0}^{t} \int_{u=0}^{s} \int_{v=0}^{u} dv \circ dW_u \circ dW_s\Big) \;=\; \frac{1}{4} t^2,$$

which agrees with Eq. (2).

References

[AG25] Abi-Jaber, E. and L. Gérard, "Signature volatility models: pricing and hedging with Fourier", preprint, 2025.

[AG25b] Abi-Jaber, E. and L. Gérard, “Hedging with memory: shallow and deep learning with signatures”, preprint, 2025.

[AGH24] Abi-Jaber, E., L. Gérard and Y. Huang, “Path-dependent processes from signatures”, 2024.

[Ger25] Abi-Jaber, E., L. Gérard and Y. Huang, “Signatures for Volatility: Pricing and Hedging”, SIAM conference talk, Miami, 2025.